

SR. NO.	PROJECT NAME	DESCRIPTION
1	Water treatment system	Implemented an IoT-based system for real-time water quality and treatment monitoring like ph measuring, tds temprature, turbodity and chlorine dosing to ensure access to clean and safe drinking water in rural and remote areas.
2	Waste segregation	Implemented waste segregation utilizing a master-slave system, where the master unit controlled segregation processes across multiple manufacturing sites, ensuring efficient resource utilization and environmental sustainability for both dry and wet streams
3	PULSE LIGHT ILLUMINATOR	The project developed a system that controls light at different frequencies This system allows independent control of light modulation at various frequencies.
4	OT CONTROL PANEL	The project aimed to deploy an IoT-enabled control panel and centralized monitoring system for real-time oversight of facility operations. Integrating IoT facilitated efficient communication, proactive maintenance, and decision-making, leading to improved efficiency
5	Infection detection with cloud data sharing	Developed an infection detection system leveraging cloud-based data sharing capabilities analysis and collaboration among healthcare providers for prompt and effective response to infectious disease
6	Medical air and vaccume pressure monitoring	The project involved developing a system to monitor air and vacuum pressure in medical facilities. This system ensured precise control and maintenance of pressure levels critical for various medical procedures, such as surgeries and sterile environments

7	Industrial automation for measuring production efficiency	The project focused on implementing industrial automation to measure production efficiency and reduce costs. By integrating sensors and data analytics, it enabled real-time monitoring of manufacturing processes. This allowed for identification of bottlenecks, optimization of workflows, and reduction of wastage. As a result, the system enhanced productivity, minimized downtime, and improved overall operational efficiency leading to significant cost savings for the organization.
8	Industrial RO controller and water atm	The project involved designing an industrial reverse osmosis (RO) controller system. This controller regulated various parameters crucial for RO operations, such as pressure, flow rate, and water quality. It offered precise control over the RO process, ensuring optimal performance and efficiency. Additionally, the controller featured advanced monitoring capabilities
9	Industrial RO controller and water atm	The project entailed developing a portable oil filling machine with online control and data monitoring capabilities. This innovative system enabled remote control and monitoring of oil filling operations through an online interface. Real-time data collection and analysis facilitated efficient management of filling processes, ensuring accuracy and consistency. Additionally, the system provided insights into performance metrics, such as fill rates and error detection, enhancing operational transparency and decision-making.
10	Plastic joining impulse welding system	The IMPULSE WELDER is heat staking system with instantaneous heating and cooling which is safe, stable and energy saving. Apply electrical current to heat tip and heat tip is heated up by Joule heat. After melting and forming thermoplastic, cool air blow solidify plastic.

11	Automatic door control system	Automatic Door Control System is an advanced solution for automating door operations with precise direction control, adjustable speed settings, and integrated safety mechanisms. It supports multiple sensor inputs (hand switch, foot switch, photo cell) and features intuitive LED indicators for monitoring. Designed for residential, commercial, and industrial applications, it ensures seamless functionality and enhanced safety.
12	Jal Purifier	JAL Purifier is an advanced water purification device designed to effectively monitor and control water quality. It provides features like Total Dissolved Solids (TDS) monitoring, automatic valve operations, and maintenance reminders to ensure optimal water quality for the user.
13	Transformer fire alarm system	This device includes a specialized fire alarm system designed for transformer safety. It continuously monitors critical parameters such as temperature, pressure, and gas levels within the transformer. In the event of a fire hazard, the system triggers an alarm within 10 seconds, ensuring rapid response to minimize damage. The fire intensity levels can be configured, allowing for customized sensitivity based on operational requirements. This system is essential for ensuring the reliability and safety.

14	Transformer oil dryout system	The Transformer Oil Dryout System removes moisture from transformer oil to ensure efficient performance and longevity. It uses water activity sensors to measure moisture content (ppm) at the inlet and outlet. Solenoid valves, a motor, and an HMI control the oil flow and monitor critical parameters like pressure, temperature, and moisture. If moisture exceeds set thresholds, the system triggers alarms and adjusts operations to maintain optimal conditions. In case of low moisture, the system enters idle mode, while continuous operation occurs if moisture levels are within acceptable limits.
15	Ultrasonic machine control system	The ultrasonic Machine Controller automates the welding process by controlling solenoid valves, ultrasonic pulses, frequency, and current parameters. The process begins with the Start button, activating a sequence of timers for delay, welding, and hold phases, ensuring precise operation. Frequency and current are monitored and adjusted during the ultrasonic pulse phase for optimal welding performance. After the cycle completes, the system resets for the next operation. The HMI interface provides real-time data on pulse status, frequency, current, timers, and cycle count, along with alerts for errors and user controls for starting or resetting the cycle.